

Program- B.TECH

Semester- III

Session- 2020-21

Course- BEE/ECA

Course code- BTEE-211

Topic- Network Theorems. (Unit-1)

Superposition Theorem

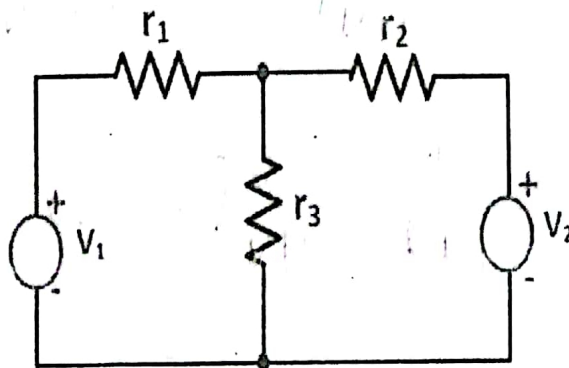
Superposition theorem states that in any linear, active, bilateral network having more than one source, the response across any element is the sum of the responses obtained from each source considered separately and all other sources are replaced by their internal resistance. The superposition theorem is used to solve the network where two or more sources are present and connected.

In other words, it can be stated as if a number of voltage or current sources are acting in a linear network, the resulting current in any branch is the algebraic sum of all the currents that would be produced in it when each source acts alone while all the other independent sources are replaced by their internal resistances.

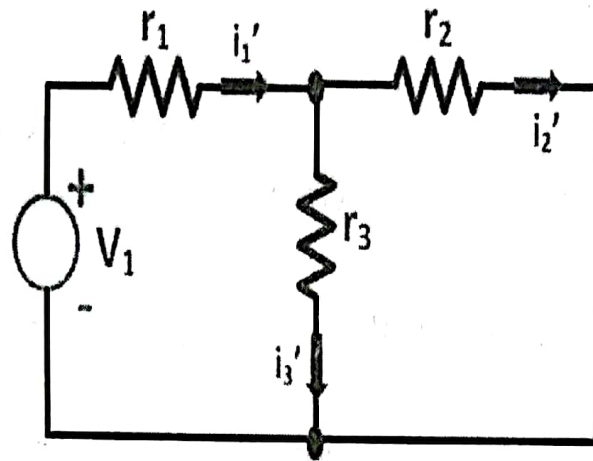
It is only applicable to the circuit which is valid for the ohm's law (i.e., for the linear circuit).

Explanation of Superposition Theorem

Let us understand the superposition theorem with the help of an example. The circuit diagram is shown below consists of two voltage sources V_1 and V_2 .



First, take the source V_1 alone and short circuit the V_2 source as shown in the circuit diagram below:



Here, the value of current flowing in each branch, i.e. i_1' , i_2' and i_3' is calculated by the following equations.

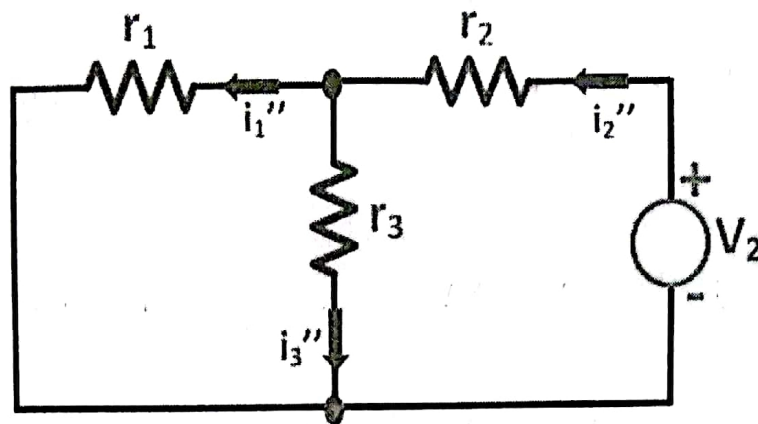
$$i_1' = \frac{V_1}{\frac{r_2 r_3}{r_2 + r_3} + r_1} \dots \dots \dots (1)$$

$$i_2' = i_1' \frac{r_3}{r_2 + r_3} \dots \dots \dots (2)$$

The difference between the above two equations gives the value of the current i_3'

$$i_3' = i_1' - i_2'$$

Now, activating the voltage source V_2 and deactivating the voltage source V_1 by short-circuiting it, find the various currents, i.e. i_1'' , i_2'' , i_3'' flowing in the circuit diagram shown below:



$$i_2'' = \frac{V_2}{\frac{r_1 r_3}{r_1 + r_3} + r_2} \quad \text{and} \quad i_1'' = i_2'' \frac{r_3}{r_1 + r_3}$$

And the value of the current i_3'' will be calculated by the equation shown below:

$$i_3'' = i_2'' - i_1''$$

As per the superposition theorem, the value of current i_1 , i_2 , i_3 is now calculated as:

$$i_3 = i_3' + i_3''$$

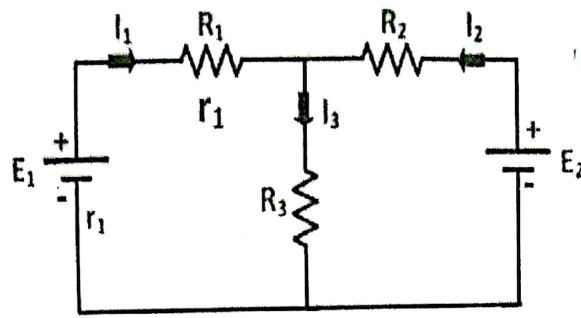
$$i_2 = i_2' - i_2''$$

$$i_1 = i_1' - i_1''$$

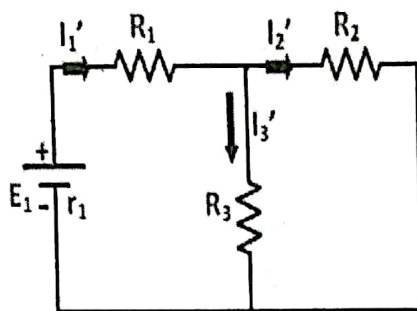
The direction of the current should be taken care of while finding the current in the various branches.

Steps for Solving network by Superposition Theorem

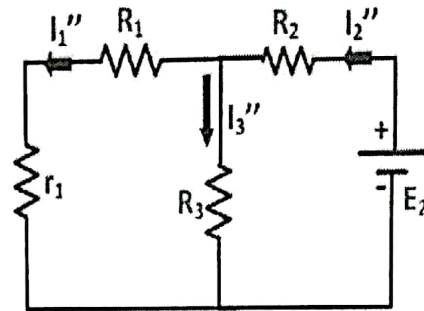
Considering the circuit diagram A, let us see the various steps to solve the superposition theorem:



Circuit Diagram A



Circuit Diagram B



Circuit Diagram C

Step 1 – Take only one independent source of voltage or current and deactivate the other sources.

Step 2 – In the circuit diagram B shown above, consider the source E_1 and replace the other source E_2 by its internal resistance. If its internal resistance is not given, then it is taken as zero and the source is short-circuited.

Step 3 – If there is a voltage source then short circuit it and if there is a current source then just open circuit it.

Step 4 – Thus, by activating one source and deactivating the other source find the current in each branch of the network. Taking the above example find the current I_1' , I_2' and I_3' .

Step 5 – Now consider the other source E_2 and replace the source E_1 by its internal resistance r_1 as shown in the circuit diagram C.

Step 6 – Determine the current in various sections, I_1'' , I_2'' and I_3'' .

Step 7 – Now to determine the net branch current utilizing the superposition theorem, add the currents obtained from each individual source for each branch.

Step 8 – If the current obtained by each branch is in the same direction then add them and if it is in the opposite direction, subtract them to obtain the net current in each branch.

The actual flow of current in the circuit C will be given by the equations shown below:

$$I_1 = I'_1 - I''_1$$

$$I_2 = I'_2 - I''_2$$

$$I_3 = I'_3 - I''_3$$

Thus, in this way, we can solve superposition theorem.